Mock Paper 1 — Computer Systems (H446/01)

Time allowed: 2 hours 30 minutes · Total: 140 marks · Answer all questions.

Instructions: Show all working on calculations and logic questions. Mark against the mark scheme at the end.

AO key: **AO1** = knowledge & understanding · **AO2** = application · **AO3** = analysis/evaluation/design.

Running mark tally is shown beside each question. Whole-paper total is verified at the very end.

Section A / questions

Question 1 — Processor architecture and the FDE cycle *(8 marks)*

A CPU uses the Von Neumann architecture and contains the registers PC, MAR, MDR, CIR and ACC.

1(a) State the role of the **Program Counter (PC)** and the **Memory Address Register (MAR)** during the fetch stage. **[3]**

1(b) Describe **three** factors, other than clock speed, that affect the performance of a processor. **[3]**

1(c) Explain **one** advantage of a Harvard architecture over a Von Neumann architecture for an embedded system. **[2]**

(Running total: 8)

Question 2 — Storage devices *(7 marks)*

2(a) State **two** characteristics of a solid-state drive (SSD) that make it suitable for use in a laptop. **[2]**

2(b) Compare optical storage with magnetic storage, giving **three** distinct points of comparison. **[3]**

2(c) A digital camera stores photographs on a removable memory card. State the **type** of storage technology used and justify why it is appropriate. **[2]**

(Running total: 15)

Question 3 — Operating systems *(9 marks)*

3(a) Explain how a **paging** memory-management technique allows a program larger than physical RAM to run. **[4]**

3(b) Describe the purpose of an **interrupt** and explain the role of the **interrupt service routine (ISR)**. **[3]**

3(c) State **one** difference between a **multi-tasking** and a **multi-user** operating system. **[2]**

(Running total: 24)

Question 4 — Translators and the compilation process *(9 marks)*

4(a) State **three** differences between a **compiler** and an **interpreter**. **[3]**

4(b) The compilation of source code passes through four stages: lexical analysis, syntax analysis, code generation and optimisation. Describe what happens during **lexical analysis** and **syntax analysis**. **[4]**

4(c) Explain why a programmer might choose to distribute software as **bytecode** for a virtual machine rather than as native machine code. **[2]**

(Running total: 33)

Question 5 — Software development methodologies *(7 marks)*

5(a) A start-up is building a mobile app where the client's requirements are expected to change frequently. Justify why an **Agile** methodology would be more suitable than the **Waterfall** model. **[4]**

5(b) Describe **three** features of the **Extreme Programming (XP)** methodology. **[3]**

(Running total: 40)

Question 6 — Object-oriented programming *(9 marks)*

A program models library members. A base class `Member` has the attributes `name` and `memberID`. A subclass `StudentMember` adds the attribute `course` and overrides the method `borrowLimit()`.

6(a) Explain what is meant by **inheritance** and **encapsulation**, using this scenario to illustrate each. **[3]**

6(b) Write the class definition for `Member` in pseudocode (or a language of your choice), including a constructor and a `getDetails()` method. Use private attributes with public accessor methods. **[4]**

6(c) Explain what is meant by **polymorphism** and give an example using `borrowLimit()`. **[2]**

(Running total: 49)

Question 7 — Assembly language and addressing modes *(8 marks)*

A processor uses the Little Man Computer (LMC)-style instruction set. Consider this program:

```
        INP
        STA  num
loop    LDA  total
        ADD  num
        STA  total
        LDA  num
        SUB  one
        STA  num
        BRP  loop
        LDA  total
        OUT
        HLT
num     DAT
total   DAT  0
one     DAT  1
```

7(a) Trace the program for an input of **3**. State the final value output and explain in one sentence what the program computes. **[4]**

7(b) Explain the difference between **immediate** and **direct (absolute)** addressing, and state **one** reason a processor also provides **indexed** addressing. **[4]**

(Running total: 57)

Question 8 — Compression, encryption and hashing *(10 marks)*

8(a) Explain the difference between **lossy** and **lossless** compression, giving **one** suitable use case for each. **[3]**

8(b) Describe how **run-length encoding (RLE)** compresses data and state **one** type of data for which it is ineffective. **[3]**

8(c) Explain how **symmetric** encryption differs from **asymmetric** encryption, and describe how asymmetric encryption can be used to create a **digital signature**. **[4]**

(Running total: 67)

Question 9 — Databases and SQL *(10 marks)*

A `Booking` table and a `Customer` table store data for a car-hire company:

```
Customer(CustomerID, Surname, City)
Booking(BookingID, CustomerID, CarReg, StartDate, Days)
```

9(a) State what is meant by a **primary key** and a **foreign key**, identifying one of each from the tables above. **[2]**

9(b) Write an SQL statement to list the `Surname` and `City` of every customer who lives in `'Leeds'`, sorted alphabetically by `Surname`. **[3]**

9(c) Write an SQL statement that lists the `Surname` of each customer together with the `CarReg` and `StartDate` of all their bookings of **5 or more** days. **[5]**

(Running total: 77)

Question 10 — Networks and TCP/IP *(10 marks)*

10(a) Describe the role of each of the four layers of the **TCP/IP stack**. **[3]**

10(b) Explain how the **Domain Name System (DNS)** resolves a URL into an IP address, referring to the hierarchy of DNS servers. **[4]**

10(c) Explain the difference between **circuit switching** and **packet switching**, and state why packet switching is used on the internet. **[3]**

(Running total: 87)

Question 11 — Number representation *(12 marks)*

11(a) Convert the denary number **−42** into **8-bit two's complement**. Show your working. **[3]**

11(b) A floating-point number uses an **8-bit mantissa** and a **4-bit exponent**, both in two's complement, with an assumed binary point after the first mantissa bit. The stored value is:

```
mantissa = 0101 1000   exponent = 0011
```

Calculate the denary value this represents. Show your working, and state whether the number is **normalised**, justifying your answer. **[5]**

11(c) Perform the binary addition `0110 1101 + 0101 0011` in 8 bits. State the 8-bit result and explain whether **overflow** has occurred if the values are interpreted as signed two's complement. **[4]**

(Running total: 99)

Question 12 — Data structures *(10 marks)*

12(a) Explain the difference between a **stack** and a **queue**, and give **one** real use of each in a computer system. **[3]**

12(b) A **circular queue** is implemented in an array of size 5 with `front` and `rear` pointers. Explain **why** a circular queue is preferred to a linear queue, and describe how the `rear` pointer wraps around. **[4]**

12(c) State the **Big O** time complexity of: (i) adding an item to the top of a stack; (ii) a linear search of an unsorted list of *n* items; (iii) a binary search of a sorted list of *n* items. **[3]**

(Running total: 109)

Question 13 — Boolean algebra and logic circuits *(12 marks)*

13(a) Complete the truth table for the expression `Q = (A AND NOT B) OR (NOT A AND B)`, and state the name of the single logic gate that this expression is equivalent to. **[4]**

13(b) Using the laws of Boolean algebra, simplify the expression `Q = A∧B ∨ A∧(B ∨ C) ∨ B∧(B ∨ C)` to its simplest form. Show each step and name the law used. **[4]**

13(c) A safety system sounds an alarm (`Z = 1`) when **either** a door sensor `D` is open **and** the system is armed `A`, **or** when a panic button `P` is pressed (regardless of the other inputs). Write the Boolean expression for `Z` and draw the logic circuit using AND, OR and any other required gates. **[4]**

(Running total: 121)

Question 14 — Networking and security *(7 marks)*

14(a) Explain the difference between a **LAN** and a **WAN**, referring to ownership and geographical scale. **[2]**

14(b) Describe how a **firewall** using **packet filtering** helps protect a network. **[3]**

14(c) Explain why a **star** network topology is more robust than a **bus** topology when a single connection fails. **[2]**

(Running total: 128)

Question 15 — Extended response: legal, moral and ethical issues *(12 marks)*

A large retailer introduces an **AI-driven facial-recognition system** in its stores. The system scans the faces of every customer who enters, matches them against a database of known shoplifters, and alerts security staff in real time. Camera footage and biometric data are stored on the company's servers for 12 months.

15 Discuss the **legal, ethical and privacy issues** raised by this system. In your answer you should consider the interests of customers, the retailer and society, and reach a justified conclusion about whether the system should be deployed. **[12]**

(Running total: 140)

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